

Project - High Level Design

On

Marketing Content Generator

Generative - AI

Institution Name: Medicaps University – Datagami Skill Based Course

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1. Introduction

1.1 Scope of the Document

This High-Level Design (HLD) document describes the architecture, workflow, and major system components of the Marketing Content Generator, a specialized Generative AI-based application built for marketing professionals.

The scope of this document includes:

- Overall system architecture and major modules
- Data flow and processing stages
- Integration with LLM APIs and Vector Databases
- Design considerations for scalability, consistency, and auditability
- Key non-functional requirements such as performance and security

This document acts as a blueprint for developers and stakeholders before moving into detailed implementation and low-level design.

1.2 Intended Audience

This HLD document is intended for:

- Software Developers: To understand component interactions and system responsibilities
- Data Engineers: For managing embeddings, vector search, and storage design

- ML Engineers: For LLM integration, prompt optimization, and RAG pipeline

- Project Managers: For tracking deliverables and ensuring scope alignment
- QA/Test Teams: For understanding expected workflows and validation points
- Academic Reviewers: For evaluating architecture and design completeness

1.3 System Overview

The Marketing Content Generator is a GenAI-powered tool that helps marketing teams create professional, high-quality content quickly.

The system transforms basic inputs such as:

- Topic
- Audience
- Tone
- Marketing goal

into structured outputs such as:

- Ad copy
- Social media captions
- Product descriptions

The system uses:

- Prompt Engineering to maintain consistent tone and structure
- Vector Database Retrieval (RAG) to include industry-standard terminology
- LLM API Integration to generate final content output

The goal is to reduce manual drafting effort while improving content quality and consistency.

2. System Design

2.1 Application Design

The application follows a modular, service-based architecture consisting of:

1. Presentation Layer (Frontend/UI)

- Allows marketers to input requirements
- Displays generated content
- Provides options for regeneration and feedback

2. Backend Application Layer

- Manages request validation
- Builds prompts
- Integrates retrieval + LLM pipeline

3. AI Processing Layer

- Embedding generation
- Context retrieval using Vector DB
- LLM-based content generation

4. Data Storage Layer

- Structured database for logs and history
- Vector database for semantic search and retrieval

This layered approach ensures scalability, maintainability, and clear separation of responsibilities.

2.2 Process Flow

The high-level workflow is as follows:

1. User submits content request (topic, tone, type)
2. Backend validates input fields
3. Prompt Engineering module generates a structured prompt
4. System creates embeddings of the topic/query
5. Vector DB retrieves relevant marketing examples or brand guidelines
6. Retrieved context is appended to the final prompt

7. LLM API generates professional marketing content
8. Output is formatted and structured
9. Final content is returned to user
10. All activity is logged for auditing and improvement

2.3 Information Flow

The information flow ensures proper movement of data across modules:

- User Input → Backend: Raw topic and content requirements
- Backend → Prompt Engine: Conversion into structured prompt templates
- Prompt Engine → Vector DB: Query embeddings for context retrieval
- Vector DB → Backend: Relevant documents/snippets returned
- Backend → LLM API: Final enriched prompt sent for generation
- LLM API → Backend: Generated marketing output returned
- Backend → User: Final formatted content displayed
- Backend → Database: Logs, history, and feedback stored

2.4 Components Design

1. User Input Module

- Collects marketing topic and preferences
- Provides dropdown options for tone and content type
- Ensures required fields are completed

2. Prompt Engineering Module

- Builds structured prompts for consistent output
- Applies formatting rules (headline + CTA + keywords)
- Supports reusable templates for different marketing formats

3. Embedding & Vector Retrieval Module

- Converts topic/query into embeddings
- Searches Vector DB for similar marketing examples
- Helps maintain brand consistency and terminology

4. LLM Content Generator Module

- Sends enriched prompt to LLM API
- Controls creativity using parameters like temperature
- Generates marketing-ready output within token limits

5. Output Formatter Module

- Ensures content is properly structured

- Adds bullet formatting, CTA, and professional tone
- Removes unwanted repetition

6. Logging and Analytics Module

- Stores request history
- Tracks user feedback
- Helps improve prompts and model performance over time

2.5 Key Design Considerations

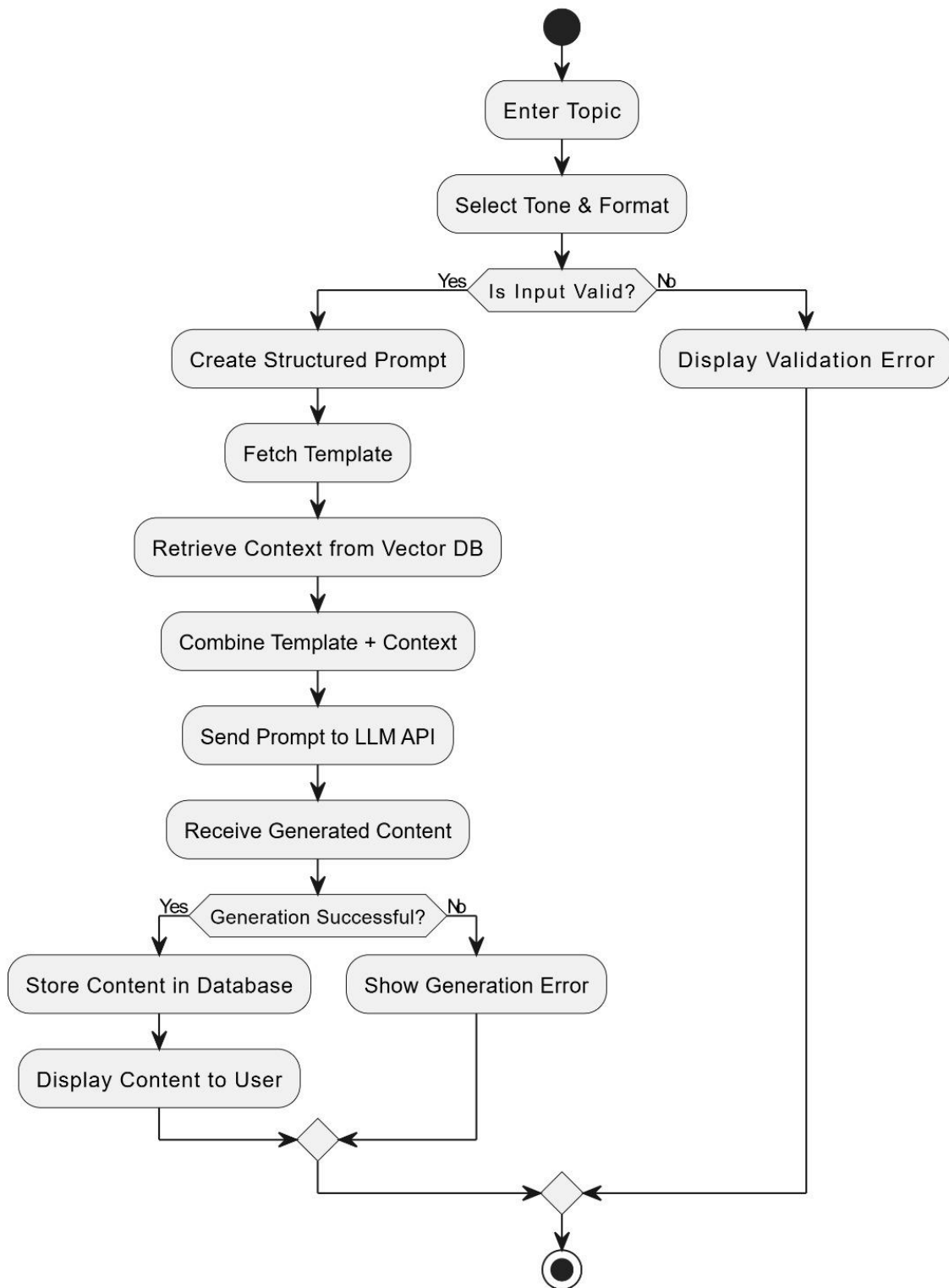
Important design factors include:

- Tone Consistency: Same brand voice across outputs
- Scalability: Support for thousands of content requests
- Auditability: Track prompt used, output generated, and user feedback
- Extensibility: Easy addition of new content categories
- Cost Optimization: Reduce repeated LLM calls using caching
- User-Friendly Workflow: Minimal steps for marketers

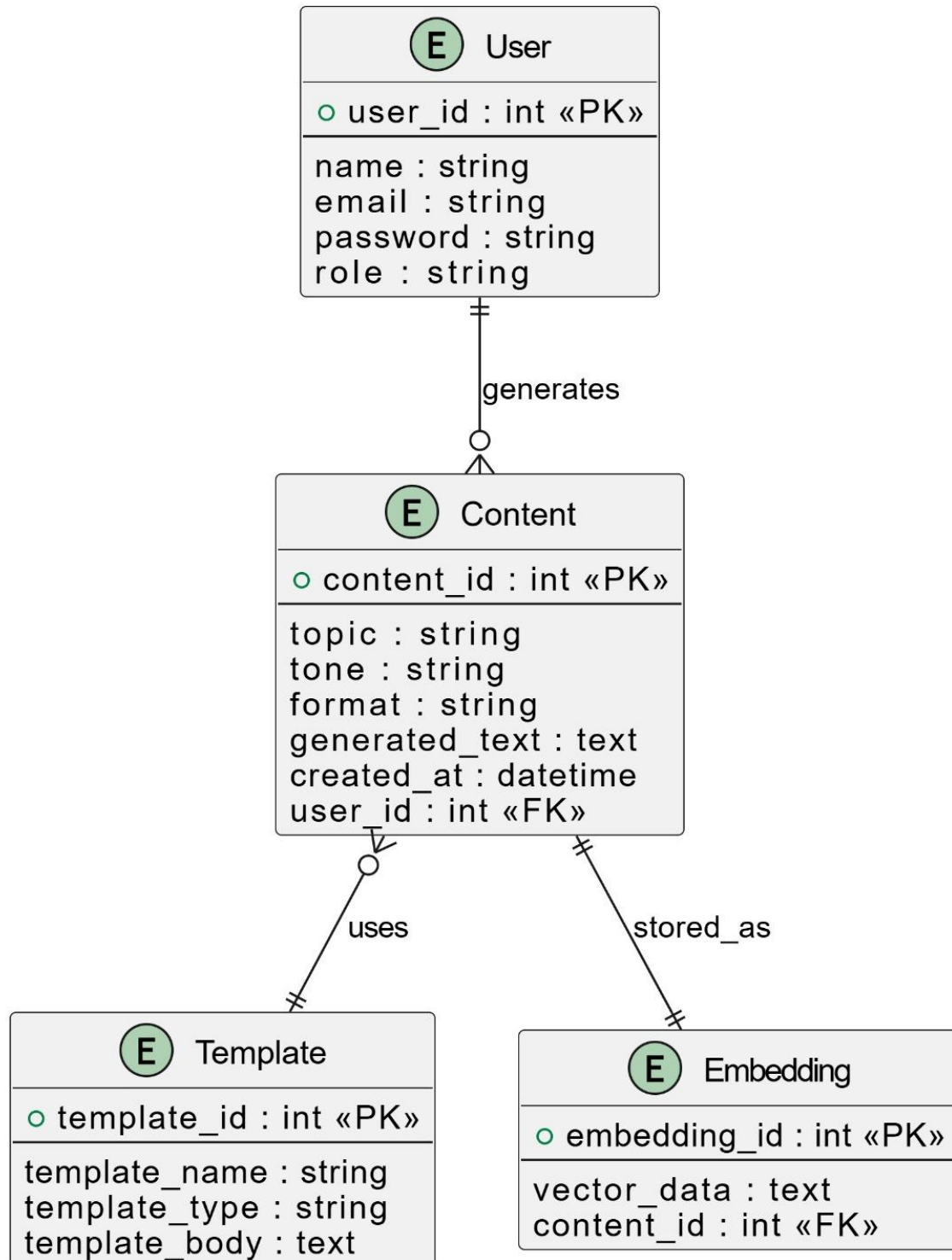
2.6 API Catalogue

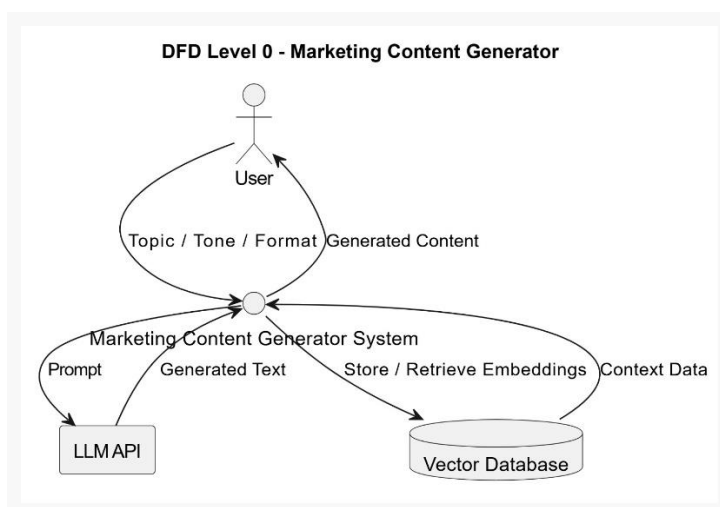
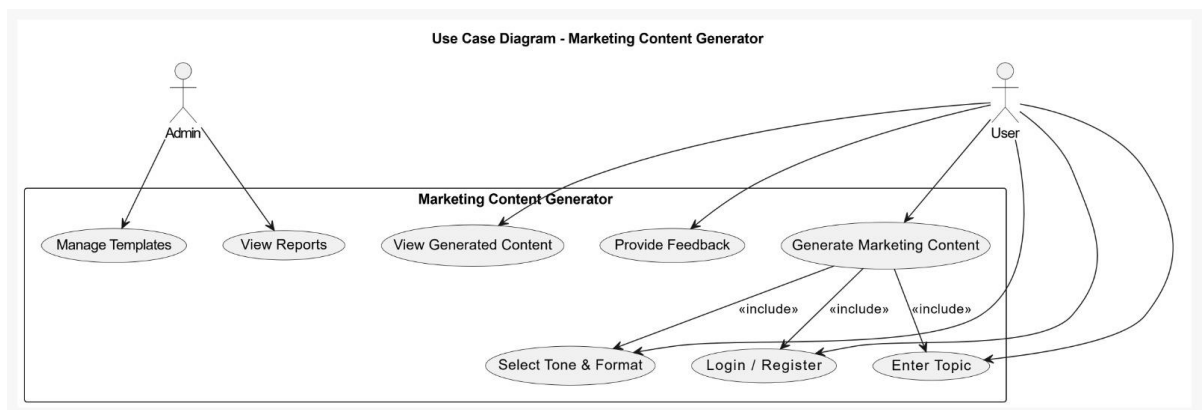
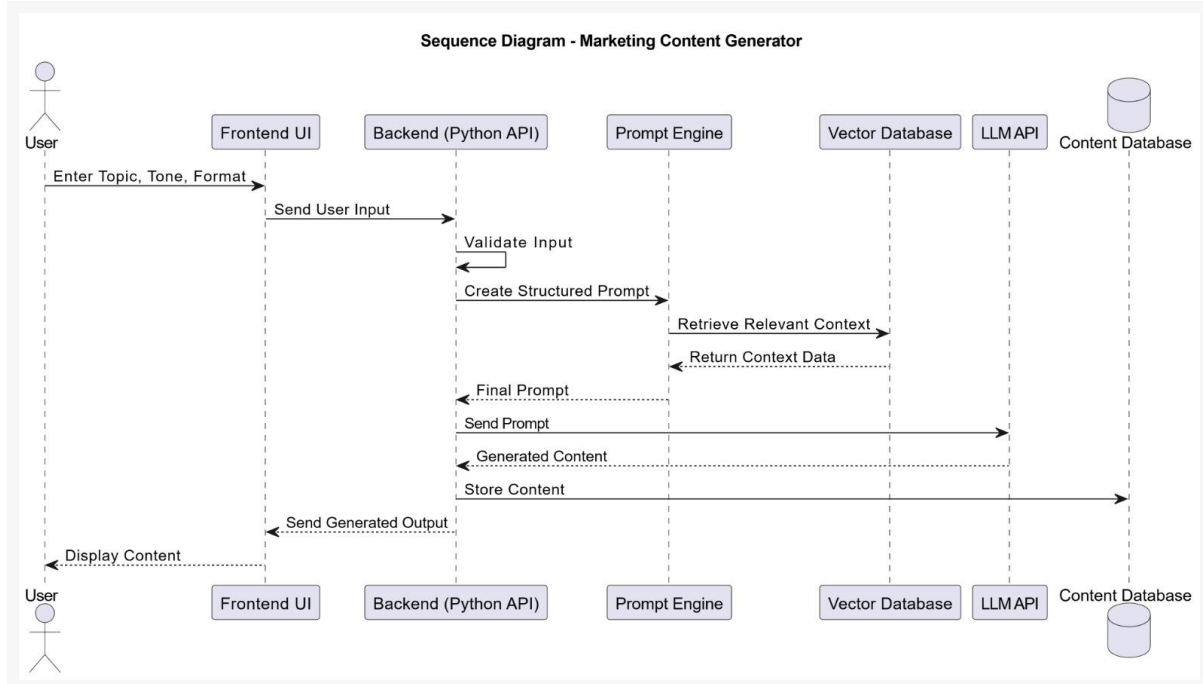
Endpoint	Method Description	
/generate-content	POST	Generates AI marketing content
/templates	GET	Lists available prompt templates
/history	GET	Retrieves user's previous generations
/feedback	POST	Stores user rating/comments
/admin/upload-context	POST	Adds new brand guidelines into Vector DB

Activity Diagram - Marketing Content Generator

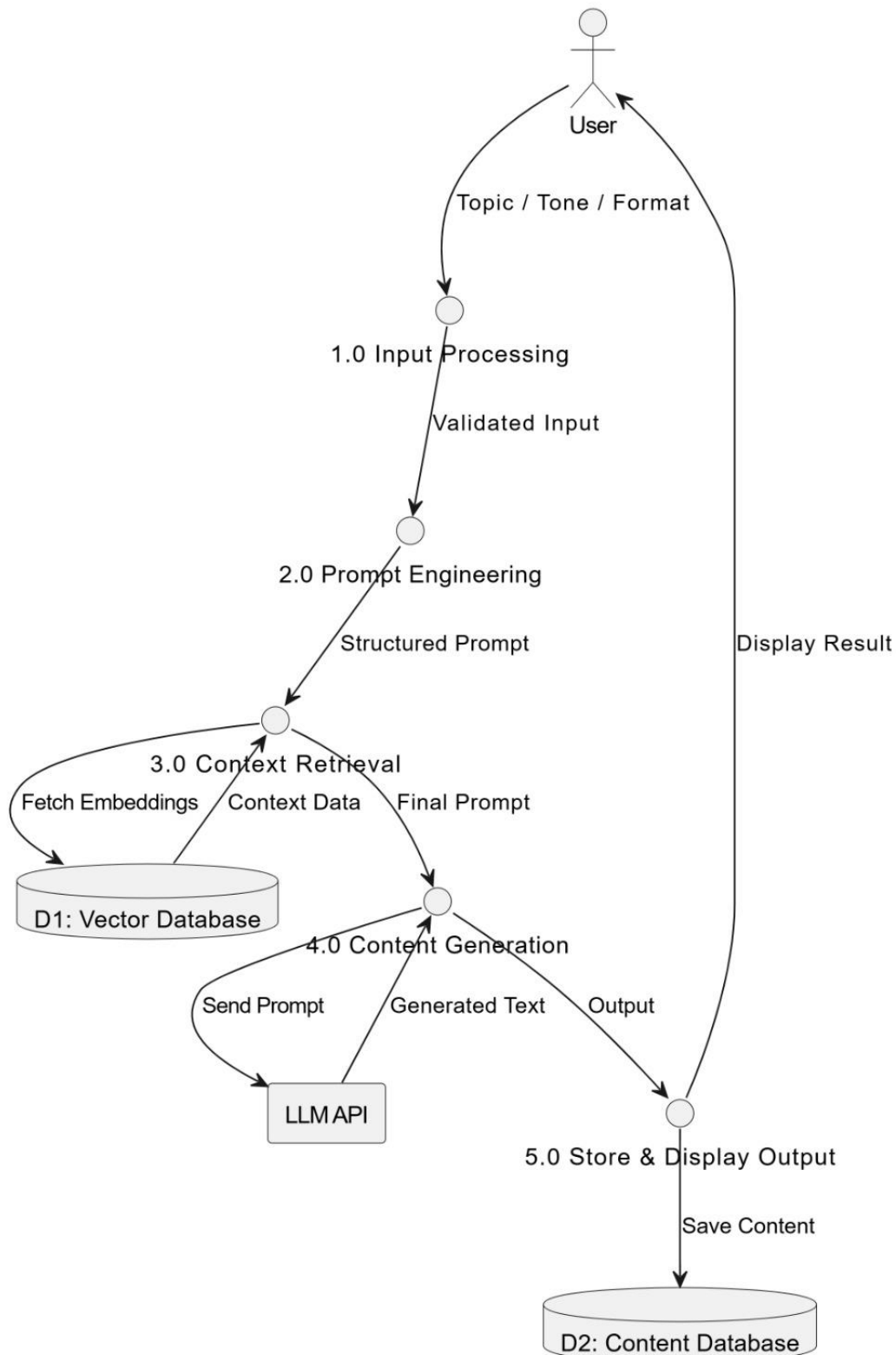


ER Diagram - Marketing Content Generator

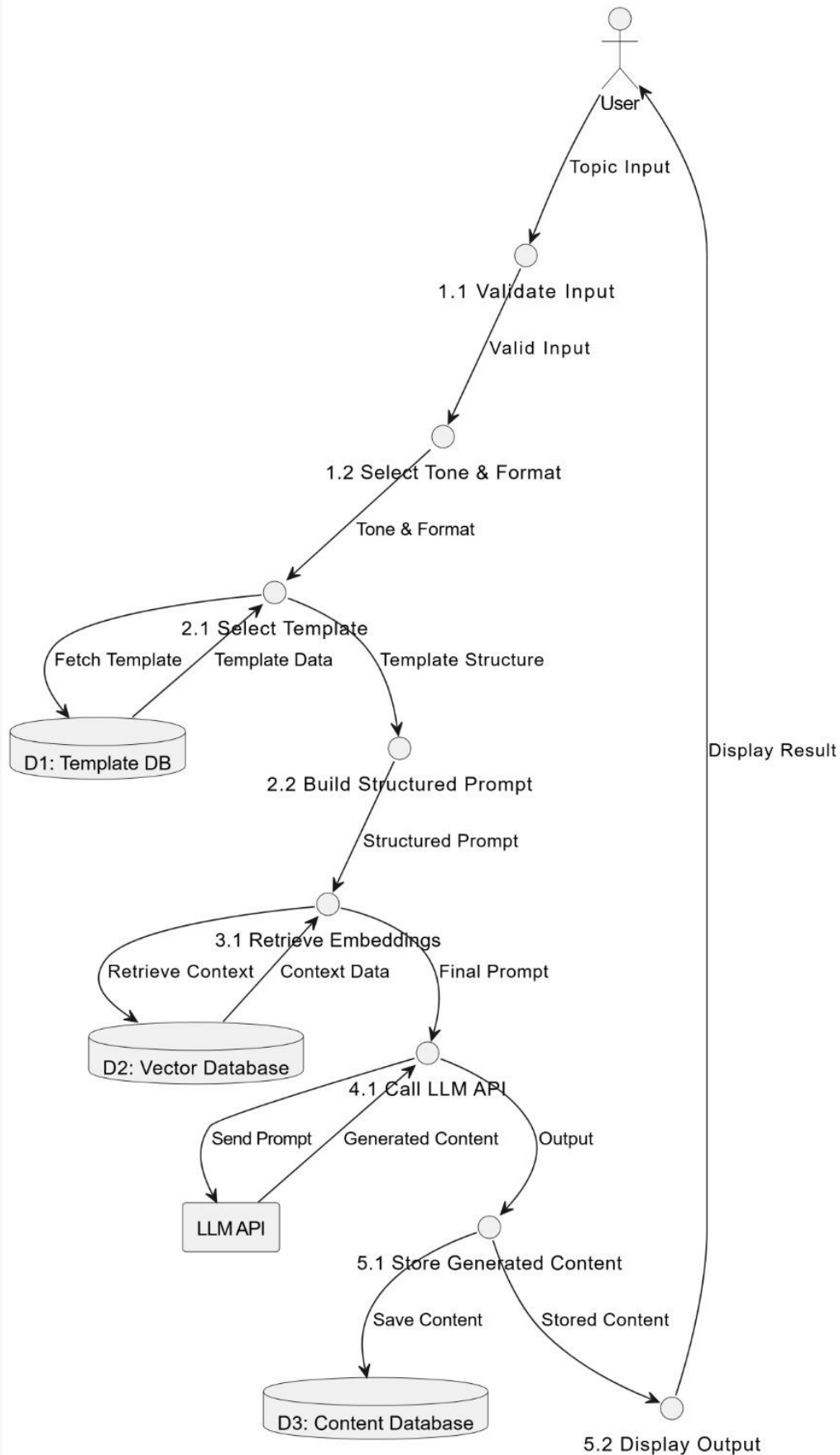




DFD Level 1 - Marketing Content Generator



DFD Level 2 - Marketing Content Generator



3. Data Design

3.1 Data Model

The system stores both structured and unstructured data.

Content Generation Log Table

- Request ID
- Topic
- Content Type
- Tone
- Prompt Used
- Retrieved Context
- Generated Output
- Timestamp
- Feedback Score

Vector Store Documents

- Brand guidelines
- Previous marketing copies

- Industry terminology documents

3.2 Data Access Mechanism

- Backend accesses structured logs via SQL queries
- Semantic retrieval happens through vector similarity search
- All access is API-controlled with authentication

3.3 Data Retention Policies

- Content logs retained for analytics and improvement
- Feedback stored to refine prompt templates
- Sensitive user data not stored unnecessarily
- Periodic cleanup of outdated drafts after defined period

3.4 Data Migration

The system supports migration strategies such as:

- Local storage → Cloud database
- Switching vector providers (FAISS → Pinecone)
- Updating embedding models over time
- Versioning prompts for backward compatibility

4. Interfaces

Interfaces provided include:

- Web dashboard for marketers
- REST APIs for third-party integration
- Admin interface for managing templates and context uploads
- Future integration with tools like Canva or Meta Ads Manager

5. State and Session Management

- Stateless REST APIs for scalability
- User authentication via JWT tokens
- Each request tracked with unique ID
- Session history stored for personalized suggestions

6. Caching

Caching improves speed and reduces LLM cost:

- Frequently used prompt templates cached
- Common topic embeddings cached
- Vector search results cached for repeated queries
- Output reuse for minor regenerations

7. Non-Functional Requirements

7.1 Security Aspects

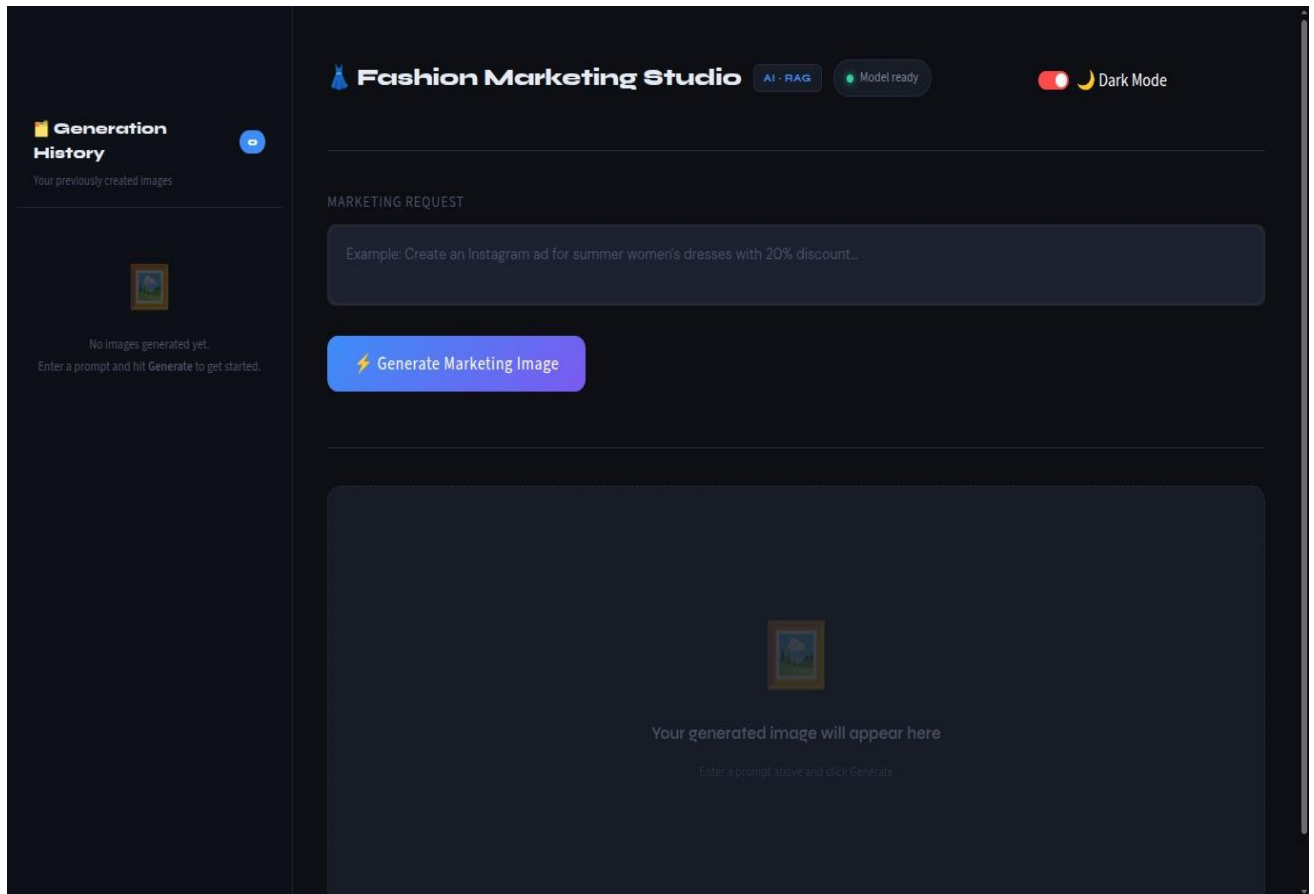
- Secure API authentication and authorization

- Encryption for stored logs
 - Input validation to prevent prompt injection
 - Role-based access for admin functions
 - Compliance with responsible AI usage policies
- ## 7.2 Performance Aspects
- Content generation response time < 5 seconds
 - Efficient vector retrieval (< 200ms)
 - Horizontal scaling using container deployment
 - Monitoring API failures and rate limits
 - Load balancing for high traffic

8. References

- OpenAI API Documentation
- Gemini API Documentation
- Pinecone / FAISS Documentation
- FastAPI Framework Guide
- Prompt Engineering Best Practices Research Papers
- Retrieval-Augmented Generation (RAG) Concepts

9. Screenshots :



MARKETING REQUEST

create an instagram ad of leather jacket for men include brand name fire on jacket with 40% discount.

⚡ Generate Marketing Image

✓ Image generated successfully!



Download Image

Generation History

Your previously created images

FEB 20, 2026



create an instagram ad of leather jacket for men include brand name fire on jacket with 40% discount.

02:43 PM

Clear All History

PROMPT USED

create an instagram ad of leather jacket for men include brand name fire on jacket with 40% discount.

RECENT GENERATIONS

